ALVN Manikantha Sudarshan

Full Stack Developer · AI / ML · Java · Spring Boot · React · Python

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PROFESSIONAL SUMMARY

M.Tech (CSE) student who builds full-stack and AI-powered applications end to end with Java, Spring Boot, React and Python. Built a Retrieval-Augmented Generation (RAG) tutor on PostgreSQL + pgvector, a deep-reinforcement-learning cyber defense platform with explainable AI, and a deep-learning medical imaging app. Seeking a software engineering role to ship reliable products and grow in applied AI and backend engineering.

EDUCATION

M.Tech, Computer Science and Engineering – Amrita Vishwa Vidyapeetham, Bengaluru — CGPA 8.1 | 2025 – 2027

B.Tech, Computer Science and Engineering – KL University, Vijayawada — CGPA 8.92 | 2020 – 2024

Intermediate (MPC) – Sree Chaitanya Junior College, Kavali — CGPA 9.57 | 2018 – 2020

TECHNICAL SKILLS

Languages	Java, Python, JavaScript (ES6+), SQL, Dart
Backend	Spring Boot, FastAPI, Express.js, REST APIs, JWT Authentication
Frontend & Mobile	React.js, Redux, Tailwind CSS, HTML5, CSS3, Flutter
AI / ML	RAG, LLMs, Vector Embeddings, Deep Reinforcement Learning, TensorFlow, PyTorch, Vision Transformers, SHAP / Explainable AI
Databases	PostgreSQL (pgvector), MySQL, MongoDB, Firebase
Tools	Git, GitHub, Docker

PROJECTS

LearnMate – AI-Powered RAG Tutor FEATURED

Project page

- Built a full-stack AI learning assistant where users upload documents and ask context-aware questions answered through **Retrieval-Augmented Generation (RAG)**.
- Implemented the ingestion pipeline – document parsing, automatic text chunking, embedding generation and semantic retrieval with **pgvector on PostgreSQL** – so answers stay grounded in the uploaded content.
- Integrated LLM response generation with the retrieval pipeline, supporting subject-aware learning, multiple difficulty levels and conversational memory.
- Secured the platform with **JWT role-based authentication**: administrators manage document uploads while learners use a conversational chat interface.

Tech: Java, Spring Boot, ReactJS, PostgreSQL, pgvector, REST APIs, JWT, LLMs, Vector Embeddings

AI-Based Adaptive Cyber Defense Simulation Platform (Deep RL)

Live: cyberdefense-beta.vercel.app

- Developed a real-time intrusion detection and adaptive defense platform using a **Dueling Double DQN** agent that classifies live network traffic into ALLOW, THROTTLE, REDIRECT and BLOCK actions.
- Integrated **SHAP explainable AI** to produce feature-level explanations for every security decision, improving transparency and trust.
- Designed and deployed the full stack – React, Express.js, FastAPI, MongoDB and JWT – for live threat monitoring, event logging and secure user management.

Tech: Python, Deep Reinforcement Learning, SHAP, React, Express.js, FastAPI, MongoDB, JWT

AI-Powered Cardiomegaly Detection Mobile Application

- Built a cross-platform **Flutter** app that detects cardiomegaly from chest X-ray images using deep learning (Vision Transformers).
- Developed a **FastAPI** inference server and integrated Firebase for authentication, cloud storage and mobile deployment.
- Integrated a locally hosted LLM to generate clinician-friendly diagnostic summaries while preserving patient privacy.

Tech: Flutter, Dart, FastAPI, Firebase, Python, TensorFlow / PyTorch, Vision Transformers, Explainable AI

School Management System

- Developed a full-stack system with ReactJS, Spring Boot and MySQL that automates student admissions, fee management and administrative operations.
- Designed RESTful APIs for student registration, fee payment tracking, balance calculation, payment history and receipt generation.

Tech: ReactJS, Java, Spring Boot, MySQL, REST APIs, Axios